

SMART AUTOMATIC TRASH BIN

A Touchless Solution For Hygienic Waste Disposal



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QUESTION #1:
WOULD YOU TOUCH
THIS?

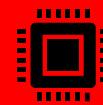




PROBLEM!



COMMERCIAL
SOLUTIONS ARE
COSTLY (\$80-\$160)



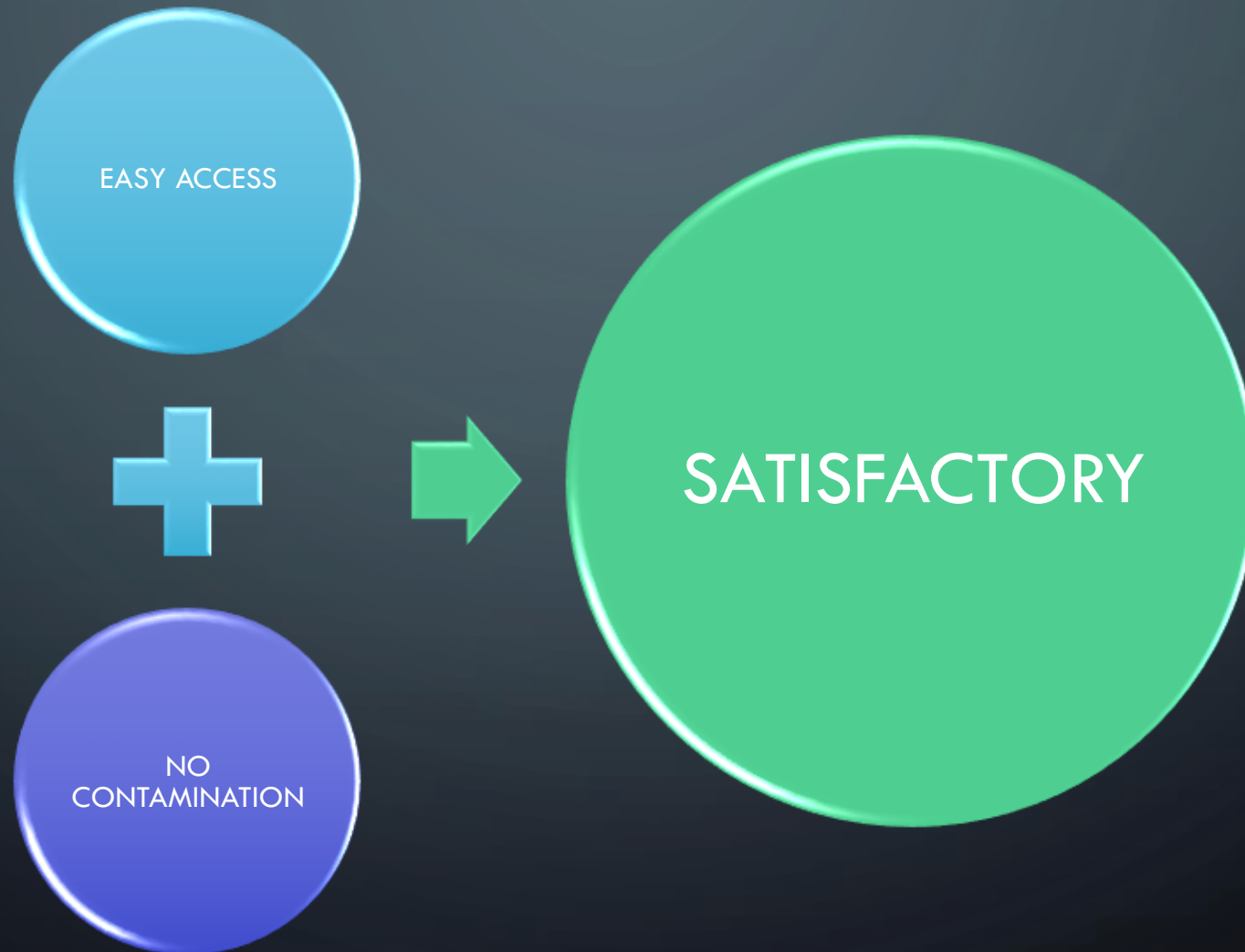
PHYSICAL CONTACT
INCREASES
CONTAMINATION RISK



LIMITED FLEXIBILITY
AND CUSTOMIZATION

- *Current solutions fail to effectively balance hygiene, cost, and accessibility.*

WHY? YOU MIGHT ASK.
THE EASY SOLUTION – AN AUTOMATIC TRASH BIN

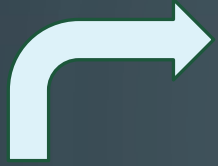


PRACTICALITY

1. Touchless automatic lid operation
2. Low-cost design (-\$60)
3. Customizable and open system



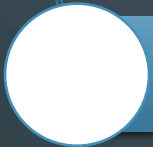
➤ Provides a practical and accessible approach to hygienic waste disposal



Key Components



Arduino UNO (Controller)



Ultrasonic Sensor (Detection)



Servo Motor (Movement)



Power Supply

- Components selected for reliability, affordability, and ease of integration.

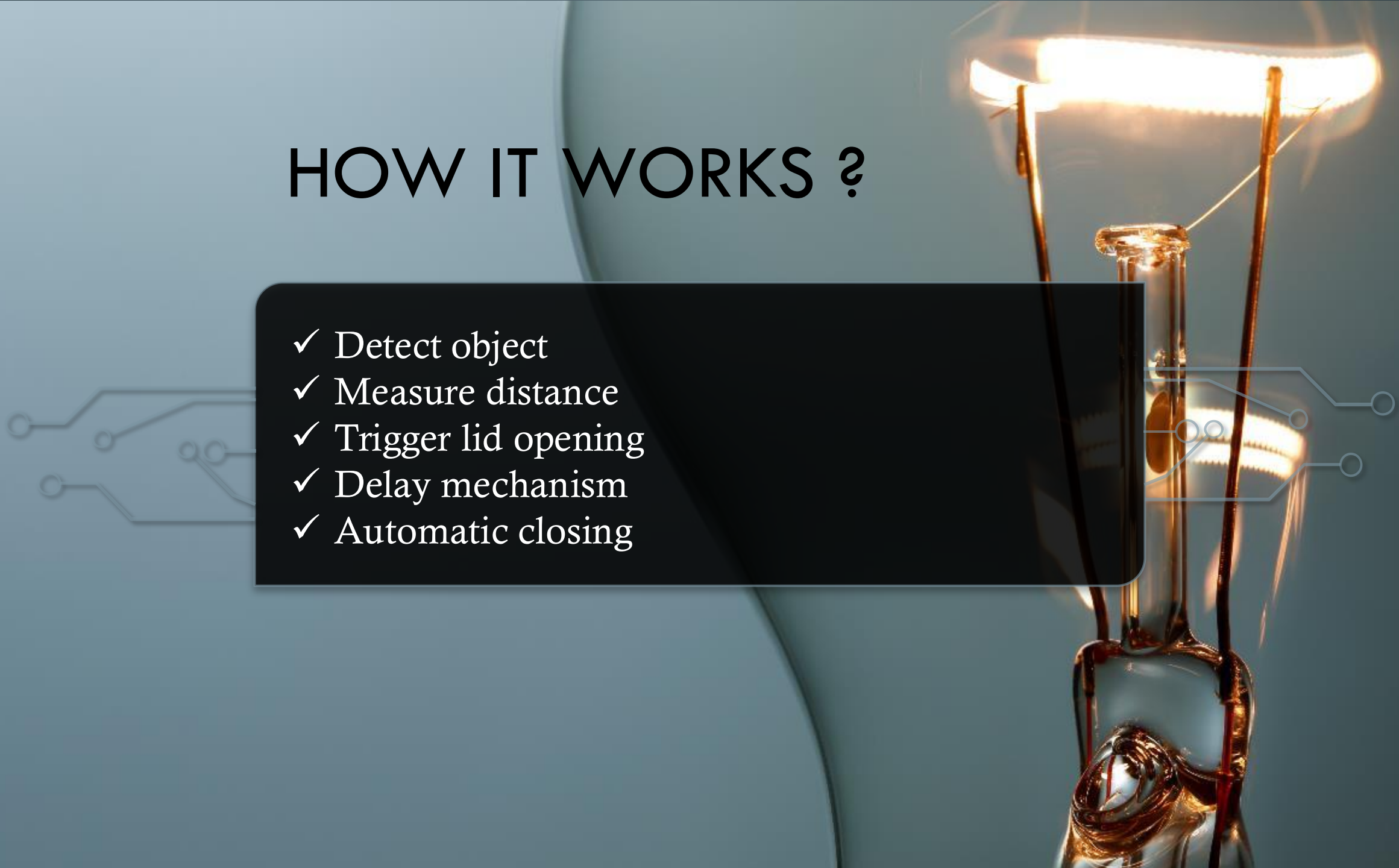
SYSTEM OVERVIEW



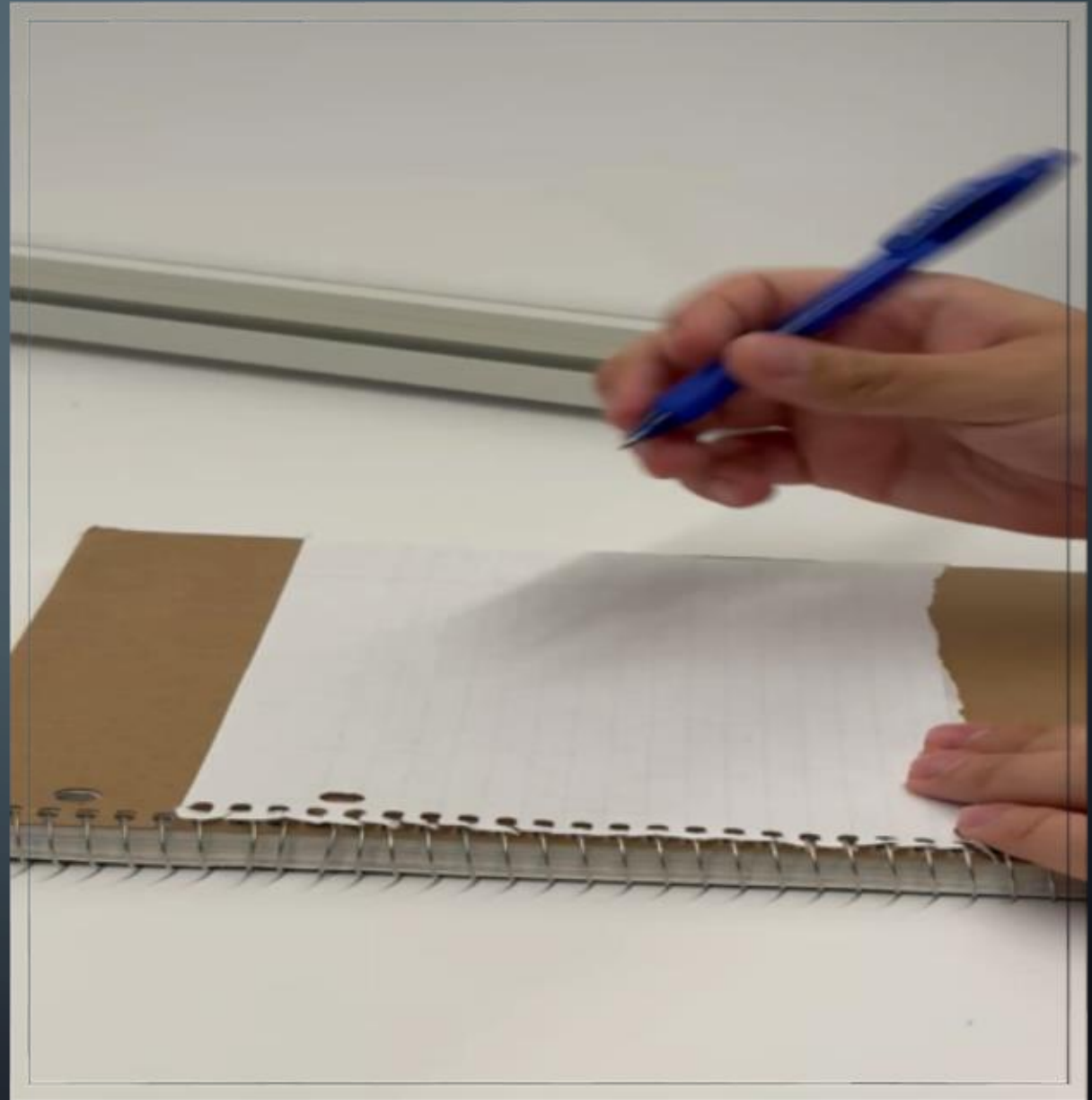
- OBJECT DETECTION TRIGGERS AUTOMATED LID OPERATION THROUGH SENSING, PROCESSING, AND ACTUATION

HOW IT WORKS ?

- ✓ Detect object
- ✓ Measure distance
- ✓ Trigger lid opening
- ✓ Delay mechanism
- ✓ Automatic closing



SYSTEM IN ACTION (DEMO)



TESTING & PERFORMANCE

1. Detection Range: 20-30 cm
2. Response Time: ~1-2 seconds
3. Consistent Operation
4. Minimal False Triggering



- Demonstrates reliable performance under repeated testing conditions

CHALLENGES

🔊 Sensor noise and interference

🔄 Repeated triggering

🕒 Timing control

- Challenges addressed through calibration and system optimization



FUTURE IMPROVEMENTS

- IoT integration
- Mobile or wireless control
- Improved enclosure
- Power optimization

~ Potential for expansion into smart home and automation systems



IMPACT



Improved hygiene



Cost effective solution



Real-world application



Educational value

~ Demonstrates the practical application of embedded systems in everyday life

Conclusion



Touchless Operation



Practical solution



Reliable performance

❑ A practical and scalable solution for hygienic waste disposal

THANK YOU!

QUESTIONS?

